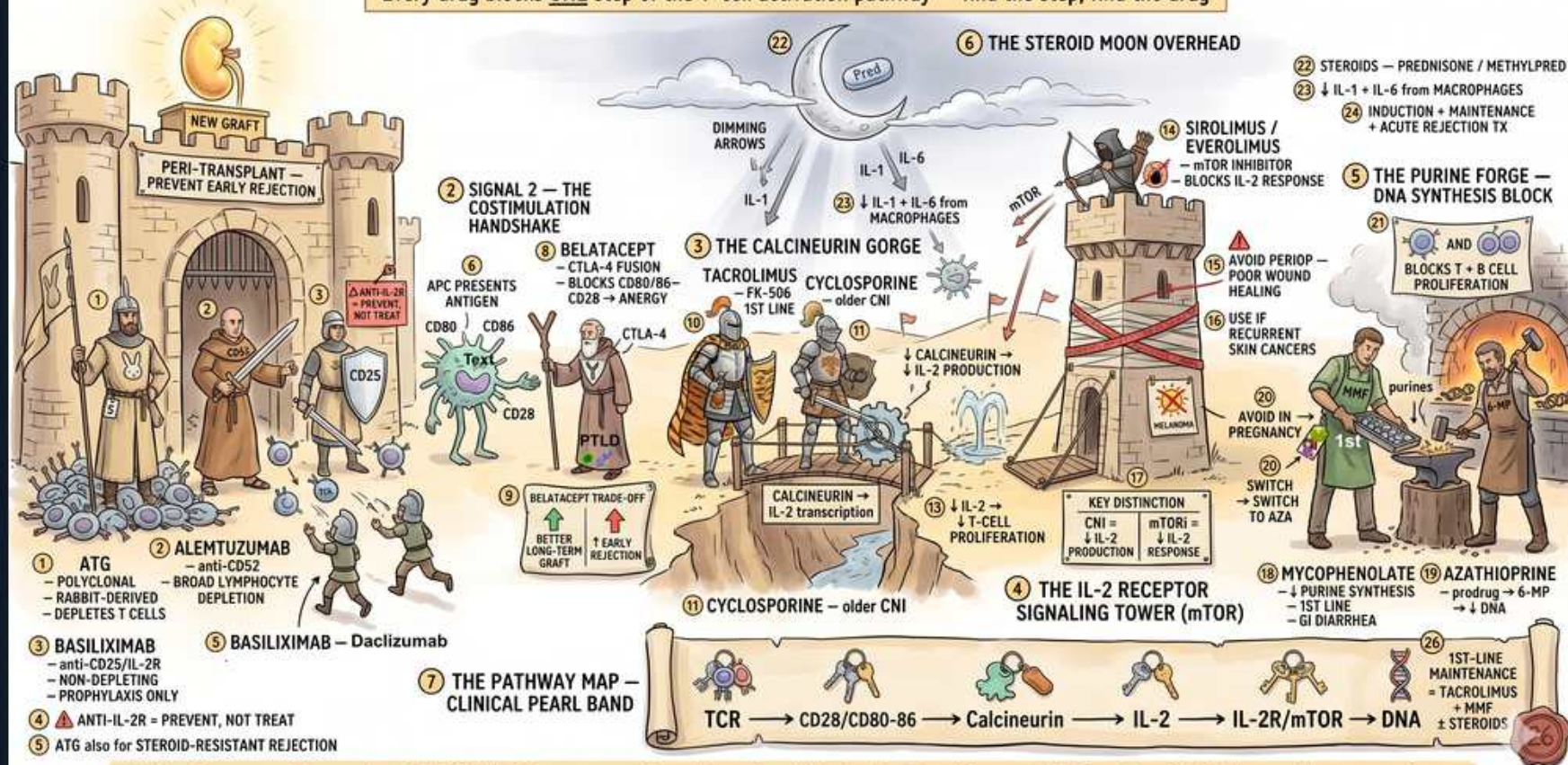


Transplant Immunosuppression — Induction Gate & Maintenance by Signal

THE T-CELL BATTLEFIELD — TRANSPLANT IMMUNOSUPPRESSION: INDUCTION GATE + MAINTENANCE BY SIGNAL

Every drug blocks ONE step of the T-cell activation pathway — find the step, find the drug



Induction = peri-transplant TRIO. Maintenance = signal-by-signal blockade. Tacrolimus + MMF ± steroids is the workhorse regimen.

1. Induction = peri-transplant, prevent early rejection

WHAT YOU SEE

New graft tower, peri-transplant induction banner

WHAT IT ENCODES

Induction agents are given at the time of transplant to blunt early, vigorous T-cell activation and prevent acute rejection in the high-risk first days. They are intense, short courses, not maintenance.

BOARD TRAP

Induction is peri-operative prophylaxis — it does not treat established rejection; that requires steroids or ATG depending on type.

2. ATG — polyclonal T-cell depletion

WHAT YOU SEE

ATG potion, rabbit-derived depleting agent

WHAT IT ENCODES

Anti-thymocyte globulin is a polyclonal depleting antibody used for induction in high-risk patients AND to TREAT steroid-resistant acute cellular rejection. Causes cytokine release, profound lymphopenia.

BOARD TRAP

ATG is the answer for STEROID-RESISTANT acute cellular rejection — basiliximab (non-depleting) does NOT treat rejection.

3. Basiliximab — IL-2R (anti-CD25), non-depleting

WHAT YOU SEE

Basiliximab shelf, daclizumab note

WHAT IT ENCODES

Basiliximab is an anti-CD25 (IL-2 receptor alpha) monoclonal used for induction in standard/low-risk patients. It is non-depleting and prevents rejection but does NOT treat it.

BOARD TRAP

Basiliximab = prevent only. Choosing basiliximab to treat ongoing rejection is wrong — use steroids first, then ATG.

4. Alemtuzumab — anti-CD52 depletion

WHAT YOU SEE

Alemtuzumab anti-CD52 depleting agent

WHAT IT ENCODES

Alemtuzumab (anti-CD52) profoundly depletes T and B lymphocytes for induction in highly sensitized patients, allowing steroid/CNI minimization but with infection risk.

BOARD TRAP

Depleting agents (ATG, alemtuzumab) carry higher infection/PTLD risk than non-depleting basiliximab — match agent to immunologic risk.

5. Signal 2 / costimulation — belatacept

WHAT YOU SEE

Costimulation handshake, belatacept CD80/86

WHAT IT ENCODES

Belatacept (CTLA4-Ig fusion) blocks the CD28–CD80/86 costimulatory Signal 2, providing CNI-free maintenance that spares nephrotoxicity. Contraindicated in EBV-seronegative recipients (PTLD risk).

BOARD TRAP

Belatacept is CONTRAINDICATED in EBV-naïve (seronegative) patients due to PTLD/CNS lymphoma risk — confirm EBV serology first.

6. Tacrolimus / cyclosporine — calcineurin inhibitors

WHAT YOU SEE

Calcineurin gorge, FK506 vs older CNI

WHAT IT ENCODES

CNIs block calcineurin → less IL-2 transcription (Signal 1/2 output), the backbone of maintenance. Tacrolimus (FK506) is first-line over cyclosporine. Both are nephrotoxic via afferent vasoconstriction.

BOARD TRAP

CNI nephrotoxicity MIMICS rejection on biopsy — check trough levels; the fix is lowering CNI, opposite to treating rejection.

7. Tacrolimus first-line vs cyclosporine

WHAT YOU SEE

Tacrolimus FK-506 1st-line label

WHAT IT ENCODES

Tacrolimus is the preferred CNI (more potent, better graft survival) but causes more post-transplant diabetes, tremor, and alopecia; cyclosporine causes gingival hyperplasia and hirsutism instead.

BOARD TRAP

Side-effect discriminator: gingival hyperplasia + hirsutism = cyclosporine; diabetes + hair LOSS + tremor = tacrolimus.

8. mTOR inhibitors — sirolimus/everolimus

WHAT YOU SEE

Sirolimus moon, blocks IL-2 response/B-cell proliferation

WHAT IT ENCODES

mTOR inhibitors block the response to IL-2 (Signal 3 / cell-cycle progression). They spare the kidney (non-nephrotoxic) but cause poor wound healing, proteinuria, mouth ulcers, hyperlipidemia.

BOARD TRAP

Sirolimus impairs WOUND HEALING — avoid in the immediate peri-operative period; classic exam pitfall after surgery.

9. Mycophenolate — purine synthesis (IMPDH)

WHAT YOU SEE

Purine forge, MMF blocks DNA synthesis

WHAT IT ENCODES

Mycophenolate mofetil inhibits inosine monophosphate dehydrogenase, selectively blocking lymphocyte purine (guanosine) synthesis. First-line antimetabolite; causes GI upset and marrow suppression. Teratogenic.

BOARD TRAP

MMF is teratogenic — counsel contraception; its GI/marrow toxicity is dose-related, distinct from azathioprine's TPMT issue.

10. Azathioprine — purine analog, TPMT

WHAT YOU SEE

Azathioprine in purine forge

WHAT IT ENCODES

Azathioprine (older antimetabolite, converted to 6-MP) blocks purine synthesis. TPMT-deficient patients get severe myelosuppression; co-administered allopurinol blocks xanthine oxidase → toxicity.

BOARD TRAP

Azathioprine + allopurinol = severe pancytopenia (blocked metabolism); MMF has largely replaced it but TPMT testing remains tested.

11. Steroids — anti-macrophage / broad

WHAT YOU SEE

Steroid sun, prednisone/methylprednisolone

WHAT IT ENCODES

Glucocorticoids suppress macrophages and broad cytokine transcription. They are part of the maintenance triple regimen and the FIRST-LINE treatment for acute cellular rejection (IV pulse methylprednisolone).

BOARD TRAP

First-line treatment of acute CELLULAR rejection is pulse steroids; ATG is reserved for steroid-resistant cellular rejection.

12. Standard triple maintenance regimen

WHAT YOU SEE

Key-distinction band: tacrolimus + MMF + steroids

WHAT IT ENCODES

The workhorse maintenance regimen is tacrolimus + mycophenolate + prednisone — a CNI, an antimetabolite, and a steroid hitting different activation signals to minimize single-drug toxicity.

BOARD TRAP

Maintenance is signal-by-signal triple therapy, NOT a single agent — induction agents (basiliximab/ATG) are not maintenance drugs.

13. Recurrent infections / malignancy = over-immunosuppression

WHAT YOU SEE

Over-immunosuppression warning, infection/cancer icons

WHAT IT ENCODES

Too much immunosuppression manifests as opportunistic infection (CMV, BK, PJP) and malignancy (skin cancer, PTLD). Management leans toward reducing the regimen and treating the trigger.

BOARD TRAP

Over-immunosuppression complications (BK, PTLD) are treated by REDUCING drugs — opposite reflex to treating rejection.

14. Boards decision / pathway band

WHAT YOU SEE

Bottom pathway map: TCR→CD28→calcineurin→IL-2→mTOR→DNA

WHAT IT ENCODES

Pathway map: TCR signal 1 → CD80/86 costimulation signal 2 (belatacept) → calcineurin (CNI) → IL-2 (basiliximab blocks receptor) → mTOR (sirolimus) → purine/DNA synthesis (MMF/AZA). Each step = a drug class.

BOARD TRAP

Match the mechanism to the step: basiliximab=IL-2 receptor, belatacept=costimulation, CNI=calcineurin, sirolimus=mTOR/IL-2 response, MMF=purine synthesis.